

Yasi Zhang

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EDUCATION

University of California, Los Angeles

Los Angeles, USA

PhD student in Statistics and Data Science

Sep 2022 – Sep 2026 (expected)

- **PhD Advisors:** Prof. Ying Nian Wu and Prof. Oscar Leong
- **Research Interests:** Generative AI, Multimodality, Representation Learning, Reinforcement Learning
- **Tools and Programming:** Python, PyTorch, Jax, Flax, Git, L^AT_EX, Linux

Fudan University

Shanghai, China

B.S. Data Science and Big Data Technology; First-Class Honor

Sep 2018 – Jun 2022

PERSONAL STATEMENT

I enjoy research from both industry and academic perspectives, having gained extensive experience by collaborating and interning at a range of leading tech companies and contributing to real-world, industry-leading products. My current research goal is to build a **mathematically principled, closed-loop, unified multimodal foundation** capable of understanding, reasoning, and generating high-dimensional data across diverse modalities. I summarized my past research and shared my vision in these [slides](#).

SELECTED INDUSTRY EXPERIENCES

ByteDance Seed, Student Researcher

Mar 2026 – Present

Manager: Dr. Peng Wang

- Worked on the scaling study of agentic generation for [Seedream](#)/ [Seedance](#).

Google Research, Student Researcher

Mar 2025 – Sep 2025

Manager: Dr. Junfeng He, Co-mentor: Dr. Michal Lukasik

- Worked on post-training for text-to-image/video/design/editing generation with [Nano Banana](#).
- First-authored one paper on text-to-image generation currently under internal embargo-period strategic review.
- First-authored [one paper \(ICML 2026\)](#) on regression-aware reinforcement learning for LLM-as-a-Judge.

Amazon AWS AI Labs, Applied Scientist Intern

Jun 2024 – Sep 2024

Mentor: Dr. Jiarui Cai

- Worked on LLM-enhanced synthetic data generation to boost object detection with Grounding-DINO.

SELECTED FIRST-AUTHORED PUBLICATIONS

Reinforcement Learning for Multi-Turn Image Editing

UCLA×Apple AIML

Advisor: Dr. Ying Nian Wu [Under Review]

Dec 2025 – Mar 2026

- Developed MT-EditFlow, a flow-matching reinforcement learning framework that optimizes reward signals specifically for multi-turn image editing.
- Mitigated compounding errors and exposure bias inherent in iterative editing processes and boosted the FLUX.1-Kontext-dev model's turn-3 performance by 6.85 points, and FLUX.2-Klein's performance by 2.89 points.

Regression-Aware Reinforcement Learning for LLM-as-a-Judge

UCLA×Google DeepMind

Advisors: Dr. Michal Lukasik [ICML 2026]

Sep 2025 – Jan 2026

- Proposed REAL, a regression-aware RL framework for LLM-as-a-Judge, employing the generalized policy gradient to maximize regression rewards by jointly optimizing Chain-of-Thought exploration and numeric score prediction.
- Achieved SOTA correlation performance across multiple evaluation benchmarks by training Mistral2-7B, Qwen3-8B, and Qwen3-32B with Verl, demonstrating strong out-of-domain generalization.

EdiVal-Agent for Multi-Turn Image Editing

UCLA×UT-Austin×Microsoft AI

Advisors: Dr. Lijuan Wang [ICLR 2026] [Chinese Blog]

May 2025 – Sep 2025

- Developed EdiVal-Agent, an object-centric evaluation framework for multi-turn image editing, using expert tools to assess instruction following, content consistency, and visual quality.
- Built EdiVal-Bench, covering 9 instruction types and 11+ SOTA editing models, outperforming zero-shot VLMs in alignment with human judgment and identifying key failure modes like exposure bias.

Restoration Score Distillation for One-Step High-Quality Generation

UCLA×Microsoft AI

Advisors: Dr. Mingyuan Zhou, Dr. Ying Nian Wu, and Dr. Oscar Leong [ICLR 2026]

Dec 2024 – May 2025

- Introduced Restoration Score Distillation, a two-phase framework that trains one-step generators directly from corrupted data by theoretically capturing the clean distribution’s eigenspace.
- Outperformed multi-step diffusion models on denoising, super-resolution, deblurring, inpainting, and multi-coil MRI in sample quality, achieving up to 30× faster sampling without clean or few-shot data.

Flow Matching Priors for Linear Inverse Image Problems

UCLA × Amazon

Advisors: Dr. Oscar Leong and Dr. Ying Nian Wu [NeurIPS 2024]

Dec 2023 – May 2024

- Developed an iterative algorithm with theoretical justification to efficiently approximate the MAP estimator for linear inverse problems using flow matching priors.
- Validated the approach through experiments on super-resolution, deblurring, inpainting, and compressed sensing, outperforming existing flow-based methods.

Attention Map Alignment with Text-to-Image Diffusion Models

UCLA

Advisor: Dr. Ying Nian Wu [ECCV 2024]

Jul 2023 – Nov 2023

- Proposed an object-conditioned energy-based attention alignment method with intensity regularization to mitigate object neglect and attribute mismatch in text-to-image diffusion models.
- Achieved top performance in multi-object and attribute alignment on both automated metrics and human evaluation, enhancing cross-attention editing with more semantically meaningful attention maps.

INVITED TALKS

1. Towards a Unified Multimodal Foundation: Evaluating · Aligning · Generating. [slides]
Qualcomm AI Research, April 2026

SELECTED PUBLICATIONS

1. [Under Submission] Jiahui Huang*, **Yasi Zhang***, Tianyu Chen, Shu Wang, Jianwen Xie, Oscar Leong, Mingyuan Zhou, Nanzhu Wang, Ying Nian Wu. MT-EditFlow: Reinforcement Learning for Multi-Turn Image Editing with Flow Matching.
2. [ICML 2026] **Yasi Zhang***, Tianyu Chen*, Mingyuan Zhou, Oscar Leong, Ying Nian Wu, Michal Lukasik. REAL: Regression-Aware Reinforcement Learning for LLM-as-a-Judge.
3. [ICLR 2026] Tianyu Chen*, **Yasi Zhang***, Zhi Zhang, Peiyu Yu, Shu Wang, Zhendong Wang, Kevin Lin, Xiaofei Wang, Zhengyuan Yang, Linjie Li, Cheng-Han Chiang, Jianwen Xie, Oscar Leong[†], Lijuan Wang[†], Ying Nian Wu[†], Mingyuan Zhou[†]. EdiVal-Agent: An Object-Centric Framework for Automated, Fine-Grained Evaluation of Multi-Turn Editing.
4. [ICLR 2026] **Yasi Zhang***, Tianyu Chen*, Zhendong Wang, Ying Nian Wu, Mingyuan Zhou[†], Oscar Leong[†]. Score Distillation Beyond Acceleration: Generative Modeling from Corrupted Data.
5. [ICLR 2026] Peiyu Yu, Dinghuai Zhang, Hengzhi He, Xiaojian Ma, Ruiyao Miao, Yifan Lu, **Yasi Zhang**, Deqian Kong, Ruiqi Gao, Jianwen Xie, Guang Cheng, Ying Nian Wu. Latent Energy-Based Odyssey: Black-Box Optimization via Expanded Exploration in the Energy-Based Latent Space.
6. [Handbook of Numerical Analysis 2026] Johannes Hertrich, Hok Shing Wong, Alexander Denker, Stanislas Ducotterd, Zhengnan Fang, Markus Haltmeier, Željko Kereta, Erich Kobler, Oscar Leong, Mohammad Sadegh Salehi, Carola-Bibiane Schönlieb, Johannes Schwab, Zakhar Shumaylov, Jeremias Sulam, German Shâma Wache, Martin Zach, **Yasi Zhang**, Matthias J. Ehrhardt, Sebastian Neumayer. Learning Regularization Functionals: A Comparative Study.
7. [Under Submission] **Yasi Zhang**, Oscar Leong. Learning Difference-of-Convex Regularizers for Inverse Problems: A Flexible Framework with Theoretical Guarantees.
8. [AISTATS 2025] Zhi Zhang, Chris Chow, **Yasi Zhang**, Yanchao Sun, Haochen Zhang, Eric Hanchen Jiang, Han Liu, Furong Huang, Yuchen Cui, Oscar Hernan Madrid Padilla. Statistical Guarantees for Lifelong Reinforcement Learning using PAC-Bayesian Theory.
9. [NeurIPS 2024] **Yasi Zhang**, Peiyu Yu, Yaxuan Zhu, Yingshan Chang, Feng Gao, Ying Nian Wu, Oscar Leong. Flow Priors for Linear Inverse Problems via Iterative Corrupted Trajectory Matching.
10. [ECCV 2024] **Yasi Zhang**, Peiyu Yu, Ying Nian Wu. Object-Conditioned Energy-Based Attention Map Alignment in Text-to-Image Diffusion Models.
11. [ECCV 2024] Yingshan Chang, **Yasi Zhang**, Jacob Zhiyuan Fang, Ying Nian Wu, Yonatan Bisk, Feng Gao. Skews in the Phenomenon Space Hinder Generalization in Text-to-Image Generation.
12. [ICASSP 2023] Hengfang Wang, **Yasi Zhang**, Xiaojun Mao, Zhonglei Wang. Transductive Matrix Completion with Calibration for Multi-Task Learning.